# Two networks.
# One control plane.

Fixed access and mobile access are managed by two disjoint toolchains, two data models and two on-call rotations. Aether puts them behind one.

10 PROTOCOLS · ONE BINARY · OPEN AGENT · MANAGED PLATFORM

THE PROBLEM

# The access network is managed twice.

Every operator runs both of these. Almost nobody runs them from one place.

### CPE / WiFi

· uCentral · TIP OpenWiFi

· TR-369 / USP

· WRP · RDK-B

· IEEE 1905.1 EasyMesh

· OpenSync

BROADBAND TEAM

### Transport / RAN

· NETCONF + YANG

· SNMP v1/v2c/v3

· gNMI streaming

· O-RAN A1 policy

· VES event collector

RAN + TRANSPORT TEAM

**When a subscriber says “my connection is bad” —**

no single system can see the home, the backhaul and the radio.

PROTOCOL COVERAGE

# Ten protocols.
# One binary.

| | |
|---|---|
| **uCentral / TIP** | WebSocket + JSON · OpenWiFi, OpenWrt |
| **TR-369 / USP** | WebSocket + MQTT · protobuf · QSDK, prplOS |
| **WRP / Xmidt** | RDK-B, zero-install via Parodus |
| **IEEE 1905.1** | EtherType 0x893a · CMDU · EasyMesh R1–R5 |
| **OpenSync** | MQTT · Plume migration path |
| **MQTT** | 3.1 / 5.0 clustered broker · USP MTP |

| | |
|---|---|
| **NETCONF / YANG** | SSH + XML · RFC 6241 / 6242 |
| **SNMP** | v1 / v2c / v3 · polling + traps |
| **gNMI** | gRPC + protobuf · OpenConfig telemetry |
| **O-RAN A1 / VES** | A1 policy → Near-RT RIC · VES collector |

ABOVE THE LINE: CPE. BELOW: TRANSPORT, BACKHAUL AND RAN.

PRECISION

# Aether is not a RIC.

# Here is what it is.

IS

— A1 client — deliver, withdraw, fetch and status policy
— VES collector — events into ONAP-style OSS
— Device.Cellular — IMEI, IMSI, RSRP, RSRQ, SINR on FWA CPE
— One data model across CPE, transport and RAN

ON THE ROADMAP

— O1 — FCAPS alongside the vendor EMS, not replacing it
— Non-RT RIC and an rApp runtime over R1
— STOMP and CoAP message transports

IS NOT

A Near-RT RIC. No E2 termination, and so no xApp runtime —
that belongs at the edge on a sub-second control loop,
not in a cloud control plane. Different product.

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The hard part was never A1. It was a data model where a TR-181 parameter from an OpenWrt box and a gNMI subscription from an aggregation switch are both first-class.

OPEN SOURCE · OPENWRT

# ac-client.

## Download. Install. Sign up.

The OpenWrt USP agent is open source under BSD 3-Clause. Three commands and the router is under management. QSDK, prplOS and RDK-B run their own native agents — nothing of ours to install there.

```
opkg install ac-client luci-app-aclient
uci set optimacs.@config[0].claim_token='...'
/etc/init.d/ac-client start
```

| | |
|---|---|
| **TR-369 / USP 1.3** | TP-469 conformance tested |
| **Post-quantum mTLS** | X25519 + ML-KEM-768, always on |
| **UCI-backed TR-181** | 47 backend operations |
| **WiFi 7 + Cellular** | EHT modes and Device.Cellular |

1,500+ OpenWrt router models. No custom firmware image.

The OpenWrt package is open. The platform is a managed service.

DEPLOYMENT

# Meet the device where it is.

No reflash. No certification cycle. No truck roll.

- **RDK-B**
  Terminates the Parodus/WRP session already running.
  No firmware change.
- **OpenWrt**
  opkg package install. 30k+ router models.
- **QSDK**
  Agent wrapping cfg80211tool and qca-hostapd.
- **prplOS**
  Speaks the TR-181 data model bus already present.
- **OpenWiFi**
  Native uCentral client.
- **Aggregation gear**
  NETCONF or gNMI. SNMP for anything older.

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**Config changes ship as JSON Patch (RFC 6902)**

~200 bytes per update, not a ~50 KB full-config push.

# Ten protocols.
# One control plane.

Built for a messy access network. Multiple CPE silicon vendors, a platform inherited from an acquisition, aggregation gear still on SNMP, and fixed wireless CPE nobody has a dashboard for.

— $0.30 / device / month at volume — no per-device AI tax
— Open source agent, managed platform, EU / US data residency
— Self-hosting licensed on Enterprise

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